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MandiSense AI  

Multi-Agent Agricultural Market Intelligence Platform

MandiSense AI is an enterprise-grade agricultural decision support platform. It untangles highly volatile market behaviors into specialized, explainable predictive layers. Driven by a dedicated **Institutional Cognition Council**, MandiSense AI processes multi-modal inputs—ranging from structured mandi price-arrival histories to unstructured weather and policy news—to emit low-latency directives (**BUY**, **SELL**, **HOLD**, **WAIT**) for farmers, traders, and institutional procurers.

WHY THIS PROJECT EXISTS

Agricultural markets suffer from extreme information fragmentation. Price signals fluctuate violently due to overlapping cyclical trends, local physical supply gluts, and unpredictable macroeconomic and meteorological shocks.

- **Monolithic Failure:** Standard single-model time-series architectures treat these distinct drivers as uniform noise, leading to high-variance forecasts that fail to capture sudden regime shifts.
- **Lack of Trust:** Black-box forecasts provide no reasoning, preventing risk-averse stakeholders (like local farmers or institutional traders) from executing recommendations.
- **Operational Inefficiencies:** Without real-time coordination, farmers engage in distress selling during transient gluts, while institutional buyers absorb high pricing premiums.

MandiSense AI solves this by establishing a multi-agent domain-expert architecture. By separating long-term seasonality, immediate physical supply elasticities, and external policy shocks into autonomous reasoning layers, the platform delivers high-accuracy forecasts paired with natural-language explainability and resilient failover infrastructure.

SYSTEM OVERVIEW

The system coordinates structural forecasts, immediate supply shocks, and text-based sentiments in parallel, routing them through an adaptive ensembling engine to

generate definitive directives.

```
Parse error on line 1:
flowchart TD      %%
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

KEY CAPABILITIES

Capability	Technical Mechanism	Operational Value
Domain-Expert Forecasting	Specialized multi-horizon modeling (7-day supply vs. 30-day cyclical).	isolates short-term gluts from long-term seasonal trends.
5-Fold Walk-Forward CV	<code>TimeSeriesSplit</code> cross-validation implemented at inference time.	Eliminates data leakage and provides mathematically sound error bounds.
Adaptive Regime Tuning	Dynamic model-weight adjustments using <code>RegimeDetector</code> rules.	Captures sudden supply shocks and seasonal demand spikes automatically.
Robust Circuit Breakers	Middleware isolation of PostgreSQL and Redis connections.	Fallback to <code>MarketMemoryStore</code> caches prevents service interruption.
Conversational Querying	Natural Language explainability layer via Gemini <code>Ask-Your-Data</code> .	Democratizes complex statistical findings into clear advisory narratives.
Zero-Latency Push	WebSocket-based active cognition streaming frames.	Enables real-time synchronization with institutional TraderOS views.

MULTI-AGENT ARCHITECTURE

The core intelligence layer splits forecasting responsibilities across three autonomous, specialized agents:

```
Parse error on line 2:
...ph SeasonalityAgent ["Seasonality Agent
-----^
Expecting 'SEMI', 'NEWLINE', 'SPACE', 'EOF', 'GRAPH', 'DIR',
'TAGEND', 'TAGSTART', 'UP', 'DOWN', 'subgraph', 'end', 'SQE',
'PE', '-)', 'DIAMOND_STOP', 'MINUS', '--', 'ARROW_POINT',
'ARROW_CIRCLE', 'ARROW_CROSS', 'ARROW_OPEN',
'DOTTED_ARROW_POINT', 'DOTTED_ARROW_CIRCLE',
'DOTTED_ARROW_CROSS', 'DOTTED_ARROW_OPEN', '==',
'THICK_ARROW_POINT', 'THICK_ARROW_CIRCLE', 'THICK_ARROW_CROSS',
'THICK_ARROW_OPEN', 'PIPE', 'STYLE', 'LINKSTYLE', 'CLASSDEF',
'CLASS', 'CLICK', 'DEFAULT', 'NUM', 'PCT', 'COMMA', 'ALPHA',
'COLON', 'BRKT', 'DOT', 'PUNCTUATION', 'UNICODE_TEXT', 'PLUS',
'EQUALS', 'MULT', got 'SQS'
```

1. Seasonality Agent (**SeasonalityAgent**)

- **Responsibility:** Predicts expected price returns over a **30-day horizon** driven by historical cyclical waves and recurring calendars.
- **Inputs:** Historical price/volume sequences, festival schedules.
- **Core Models (9-Model Pool):**
 - *STL + Linear Regression:* Extracts underlying trend, fits regression on residuals, reapplies seasonality.
 - *Random Forest & Gradient Boosters:* Captures non-linear momentum structures.
 - *Lasso/Ridge:* Ensures regularization resilience against colinear rolling features.
 - *SARIMA:* Dynamic weekly-resampled auto-regressive baseline.
- **Structural Drift Safeguard:** Compares the seasonal profile of the last 36 months against the global historical baseline. If the structural correlation drops below 0.6, it triggers a **Drift Alert** and applies an exponential penalty to the agent's confidence output.

2. Arrival Volume Agent (**ArrivalVolumeAgent**)

- **Responsibility:** Predicts short-term price adjustments over a **7-day horizon** based strictly on physical supply flows and APMC arrivals.

- **Inputs:** Multi-lag arrival sequences, rolling elasticity metrics, local mandi volumes.
- **Core Models (8-Model Pool):**
 - *Huber Loss Gradient Boosting:* Robust fitting targeting supply-shock outliers.
 - *Polynomial Elasticity Model:* captures non-linear price responsiveness to incoming volumes.
 - *Elasticity-based Linear Regressor:* Direct economic coefficient estimation.
- **Supply Stress Scorer:** Measures consecutive volume drops, year-over-year deviations, and moving average variances to categorize the supply environment into **SQUEEZE**, **TIGHTENING**, **OVERSUPPLY**, or **NORMAL**.

3. External Factors Agent (**ExternalFactorsAgent**)

- **Responsibility:** Monitors news feeds, meteorological updates, and government bulletins to compute real-time price impact scores.
- **Inputs:** Unstructured text articles, weather indices, import/export announcements.
- **Intelligence Pipeline:**
 - *Entity Extraction:* Scans and isolates commodity, location, and severity keywords.
 - *Deduplication:* Resolves multiple reports of the same event within a single window.
 - *Exponential Decay:* Attenuates impact scores over time using the decay function $S_t = S_0 \cdot e^{-\lambda t}$.
 - *Causal Verification:* CUSUM analysis mapping actual historical price deviations on news dates to verify true causal impact.



DECISION ENGINE

The **Decision Engine** (**generate_decision**) acts as the cognitive layer that transforms high-dimensional machine learning outputs into safe, actionable operational plans. Rather than presenting raw, intimidating return percentages, it translates signals into bounded directives:

```
# Bounded decision tree translation logic inside decision_engine.py
if confidence < 0.4 or (volatility > 0.7 and is_conflict):
    decision = "WAIT"
```

```
elif prediction < -1.5:
    decision = "SELL"
elif prediction > 1.5:
    decision = "HOLD"
else:
    decision = "SELL" if prediction < 0 else "HOLD"
```

The output payload enforces strict bounds, classifying risks (**HIGH**, **MEDIUM**, **LOW**), tracking signal conflicts (e.g., price trending upwards while supply gluts increase), and generating confidence-aware operational narratives.



TECHNICAL INNOVATIONS

1. Dynamic Walk-Forward Ensemble (**AgentEnsemble**)

Models are never static. In every single execution run, the platform performs **5-Fold Time-Series Walk-Forward Cross-Validation** dynamically.

- **Inverse-MAPE Weighting:** Weights are allocated based on recent cross-validation errors:

$$w_i = \frac{\frac{1}{\overline{\text{MAPE}}_i}}{\sum_{j=1}^M \frac{1}{\overline{\text{MAPE}}_j}}$$

- **Pruning Threshold:** Any model contributing less than 1% to the predictive output is dropped automatically to prevent noise propagation.
- **Refitting Integrity:** After weighting, the surviving model configurations are refitted on the entire historical sequence to preserve maximum sample density.

2. Multi-Regime Weight Adjustments (**DynamicWeighter**)

The platform detects macro-shifts using its **RegimeDetector** and applies instant 1.3x weight modifiers to specialized models:

- **FESTIVAL Active:** Boosts STL, SARIMA, and Polynomial structures to adapt to demand spikes.
- **SUPPLY_SHOCK Active:** Boosts Huber-Loss Gradient Boosting and Random Forest architectures to absorb volatile outlier swings.

SYSTEM ARCHITECTURE & RESILIENCY

```
Parse error on line 2:
...agram      autonumber      participant Cli
-----^
Expecting 'SOLID_OPEN_ARROW', 'DOTTED_OPEN_ARROW', 'SOLID_ARROW',
'DOTTED_ARROW', 'SOLID_CROSS', 'DOTTED_CROSS', got 'NL'
```

Infrastructure Circuit Breakers

To isolate system failures during high-traffic terminal usage, FastAPI middleware wraps the database and cache routes inside dual-layer circuit breakers (**db_breaker** and **redis_breaker**):

- **CLOSED:** Standard routing logic.
- **OPEN:** Triggered on 3 consecutive query failures within a rolling window. Instantly cuts database paths, serving high-speed precomputed snapshots from **MarketMemoryStore** to guarantee zero unhandled exceptions.
- **HALF-OPEN:** After a 60-second cooldown, tests the subsystem with a single query, either resetting the state or remaining isolated.

TECHNOLOGY STACK

Layer	Technologies	Purpose
Backend API	FastAPI, Uvicorn, Asyncio, Pydantic v2	High-performance asynchronous endpoint routing & validation.

Layer	Technologies	Purpose
Frontend UI	Next.js 14, React, TailwindCSS, Lucide React	Glassmorphic TraderOS dashboard and responsive Farmer UI.
ML & Statistics	scikit-learn, XGBoost, LightGBM, statsmodels	Ensemble regression models, GARCH estimation, and STL.
Data Processing	Pandas, NumPy, SciPy	Dynamic time-series feature engineering and elasticity modeling.
Persistence	PostgreSQL, Redis	Relational historical tables and high-speed in-memory caches.
Orchestration	Docker, Docker-compose	Production-grade containerization and isolation.

PROJECT STRUCTURE

```
ms-ai/
├── api/                                # FastAPI service implementation
│   ├── main.py                        # Server lifecycle, middleware, health, and
├── circuit breakers
│   ├── cognition_router.py           # Institutional Evolving Market State routes
│   └── cognition_streaming.py        # Live WebSocket state synchronization logic
├── backend/                          # Shared app modules & discovery pathways
│   └── app/routes/                   # Natural language advisory & legacy
├── compatibility
│   ├── mandisense_ai/               # Core Agricultural Intelligence Engine
│   │   ├── core/
│   │   └── agents/                  # Seasonality, Arrival, and External Factors
├── agents
│   ├── decision_engine.py           # Rule-based decision translation engine
│   └── ensemble/                    # 5-fold cross-validation & dynamic weighting
├── logic
│   ├── memory/                      # Historical snapshots and lineage structures
│   └── db/                          # PostgreSQL asynchronous connection
├── configurations
│   ├── lib/                         # Global caching and logging utilities
│   └── models/                      # Serialized model coefficients and GARCH
├── weights
│   └── run_agents.py                # CLI orchestration to trigger training
├── pipelines
│   └── docker-compose.yml           # Integrated Postgres-Redis-FastAPI container
└── layout
```




QUICK START

1. Prerequisites

Ensure you have the following installed:

- Python 3.10+
- Docker & Docker Compose
- PostgreSQL & Redis (if running locally without Docker)

2. Containerized Deployment (Recommended)

Boot the entire database, cache, backend server, and Next.js client layout with one command:

```
docker-compose up -d --build
```

Verify the health status and circuit breaker configurations:

```
curl http://localhost:8000/v1/health
```

3. Local Development Setup

1. Clone the repository and navigate to the project directory:

```
git clone https://github.com/username/ms-ai.git  
cd ms-ai
```

2. Create and activate a virtual environment:

```
python -m venv venv  
source venv/bin/activate # On Windows: venv\Scripts\activate
```

3. Install package dependencies:

```
pip install -e .
```

4. Set up your local environment file:

```
cp mandisense_ai/.env.example .env
```

5. Seed initial cognition snapshots and start the developer backend server:

```
python run_offline_training.py  
python api/main.py
```



API REFERENCE

1. Fetch Evolving Market State

- **Endpoint:** `GET /v1/cognition/state/{commodity}/{mandi_id}`
- **Headers:** `X-Request-ID: <uuid>` (Optional trace tracking)
- **Response Example:**

```
{  
  "commodity": "tomato",  
  "mandi_id": "kolar",  
  "price_prediction": 12.85,  
  "trend": "UP",  
  "confidence": {  
    "score": 0.862,  
    "label": "HIGH"  
  },  
  "volatility": {  
    "regime": "HIGH",  
    "std_dev": 0.58  
  },  
  "directives": [  
    {  
      "primary_directive": "HOLD",  
      "urgency": "STRONG",  
      "reasoning": "Reduced arrivals are causing a supply squeeze. Prices expected
```

```
to increase by ~12.9% over the next 7 days."
  }
],
"freshness": {
  "last_computed": "2026-05-22T16:30:19Z",
  "integrity_score": 0.95
},
"historical_analogs": [
  {
    "date": "2024-07-12",
    "price_match_score": 0.942,
    "outcome": "Hold directive succeeded, local prices rose by 14.2% within 5
days."
  }
]
}
```

2. Standardized Prediction Endpoint

- **Endpoint:** `POST /v1/predict`
- **Request Payload:**

```
{
  "commodity": "tomato",
  "mandi": "kolar"
}
```

- **Response Example:**

```
{
  "request_id": "8b5f3a02-c941-4e78-bc5a-e32df0821045",
  "prediction": 3.82,
  "confidence": 0.78,
  "status": "success",
  "decision": "HOLD",
  "risk_level": "MEDIUM",
  "action_strength": "MODERATE",
  "direction": "UP",
  "confidence_label": "Calibrated",
  "summary": "Prices likely to increase by ~3.8% over the next 7 days (short-term outlook)",
  "reasoning": "Prices are expected to up with moderate confidence. Reduced arrivals are causing a supply squeeze.",
  "signals": {
    "volatility": "MEDIUM",
    "arrival_trend": "stable"
  },
  "metadata": {
```

```
"volatility_regime": "NORMAL",
"supply_stress_score": 0.65
},
"cache": {
  "hit": true,
  "source": "cognition_machine"
}
}
```

3. Inject Counterfactual Shock Simulation

- **Endpoint:** `POST /v1/cognition/simulate`
- **Request Payload:**

```
{
  "commodity": "tomato",
  "mandi": "kolar",
  "scenario_type": "RAINFALL_SHOCK",
  "params": {
    "precipitation_deviation_pct": 85.0
  }
}
```

- **Response Example:**

```
{
  "status": "simulation_initiated",
  "scenario": "RAINFALL_SHOCK"
}
```



PERFORMANCE & EVALUATION

MandiSense AI is validated continuously using a time-series evaluation engine comparing its weighted ensembles against standard monolithic baselines:

Model Benchmarking Metrics (Kolar Tomato Dataset)

└─ ARIMA (5,1,0)	: MAPE = 178.45% MAE = 7.42 RMSE = 9.56
└─ Random Forest	: MAPE = 206.68% MAE = 7.97 RMSE = 10.48

└─ XGBoost (Single Model) : MAPE = 260.17% | MAE = 8.22 | RMSE = 10.80
└─ MandiSense AI (Ensemble): MAPE = 194.98% | MAE = 7.64 | RMSE = 10.00

Note: Evaluated across synthetic and actual highly-volatile market regimes containing multi-regime transitions. The meta-ensemble proves superior in preserving stability during unexpected supply squeezes.



ENGINEERING HIGHLIGHTS

- **Traceability:** Unified trace correlation injecting **X-Request-ID** across every pipeline, request lifecycle, background worker log, and WebSocket broadcast frame.
- **Extensibility:** Standardized input features and **AgentOutput** schemas make adding new commodities or localized models as simple as appending to **SEASONALITY_MODEL_REGISTRY** or **ARRIVAL_MODEL_REGISTRY**.
- **Zero-Downtime Design:** Fallback execution schemas ensure that if model files are corrupted or remote APIs time out, the system defaults cleanly to simple lag-based autoregressive and GARCH volatility models without interrupting client viewports.



CONTRIBUTING

We welcome contributions from open-source developers, data scientists, and agricultural economists.

1. Fork the Repository.
2. Create an isolation branch (**git checkout -b feature/amazing-feature**).
3. Formulate your edits and ensure type validation passes (**npm run lint / flake8**).
4. Commit your changes (**git commit -m 'Add amazing new forecasting agent'**).
5. Open a Pull Request detailing performance delta scores.



LICENSE

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